

6Ps Game

The next decade

name of the game: the 6Ps - Pun Intended

The 6Ps stand for

- Progress
- Prediction
- Problem
- Prevention
- Pathway
- ... and Pun

These appear as different card types in the game.

The *Pun* refers to Peter Diamandis' [6Ds](#) of exponentials: digitized, deceptive, disruptive, demonetized, dematerialized and democratized,

the power of ideas to transform the world is itself accelerating
(Kurzweil, Ray. *Danielle: Chronicles of a Superheroine*)

The difference between *Progress* and *Prediction* is that *Progress* is what is going on today - we take these from news sources that follow science and technology - whereas *Predictions* are major milestones that we have estimated future timeline for.

Likewise, a *Problem* is something that is present and to be solved right away. A *Prevention* is a future problem where we have a bit more time to prepare.

Pathways are ideas that players dream up that on the long-term lead towards a solution; they connect progress and problems.

Learning Objectives

- See how application of emerging technologies one hears about in the news connect to the goals humanity wants to achieve
- Learn some ideas and concepts
 - The rate of change is accelerating - change is not (the only) constant
 - Convergence
 - There is no crazy idea, only an idea whose time has not come yet
 - The best way to make a million dollars is to impact a million people
 - Money is stored energy
 - Technology is what takes something scarce and makes it abundant
 - Crowdfunding
- With great power comes great responsibility - sense making of *bright* side and *dark* side of exponential progress
 - The superpowers enabled by emerging technologies
 - Trends and predicted future timelines
 - The global problems AKA grand challenges AKA UN Sustainable Development Goals
 - Existential threats
- Imagining a better world
- Learn about pathways; how we can use technology to solve - or better yet, prevent - big, global problems.

Terminology: Rounds and turns

Turn

The game is turn based. Players proceed one by one.

Round

A round is all the players having their turns one after the other. When the turn reaches the first players again then a new round starts.

Objective of the Game

Brainstorm pathways towards solutions for the world's global problems.

It is a sort of open ended, semi-cooperative game. In other words: if we don't solve these problems, we all lose. On the other hand, if we do solve these problems then we impact a lot of people in a good way.

The costs and rewards in the game are counted in I\$ - pronounce Impact Dollar.

This represents that

the best way to make a million dollars is to impact a million people

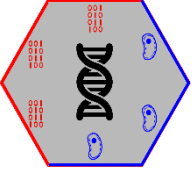
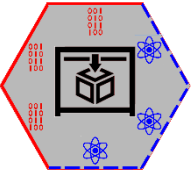
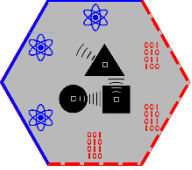
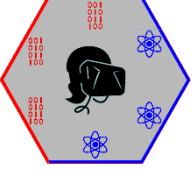
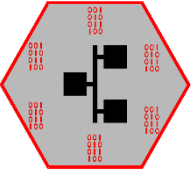
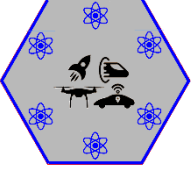


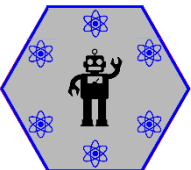
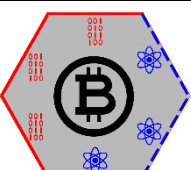
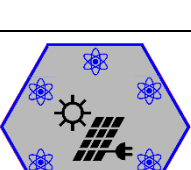
Figure 1 - Game Board

Emerging Technologies

There are eleven categories of emerging technologies – or “emTech” for short - in the game. This categorization is somewhat arbitrary. Some categories encompass a whole slew of emerging technologies. The cards are hexagon shape so that they can be placed on the table next to each other. The red and blue colors along the left and right edges show that these technologies work either in the digital world, the world of bits (red); in the real world, the world atoms (blue) or both.

Dashed borderline indicates that there is the direction cannot be incoming from that way. 3D printing and IOT are thus one-way.

	Synthetic Biology	genetic engineering, gene sequencing, gene editing	Gene sequencing goes from the real world to the world of bits: the genetic information encoded in the DNA becomes digital. Genetic engineering goes the other way: a synthetic life-form or a genetically modified organism becomes a real.
	3D printing	3D printing, additive manufacturing, nanotechnology, material science	Manufacturing starts from the imagined, represented in the digital world as information and makes it real in the world of atoms.
	IOT	internet of things - which is really networks and sensors, but since we already have networks... it's IOT and sensors	At its core IOT is sensing: measuring and capturing the real world so it becomes computable information.
	VR	virtual reality (VR), augmented reality (AR), mixed reality (MR), spatial web, metaverse	Virtual, augmented and mixed reality can be thought of as one creating a “link” between the real and the imagined, digital world – either tricking our senses completely or augmenting what we see and hear by overlaying on top of the real world the world of bits.
	Networks	5G, global broadband, quantum internet	Networks and communication move information in 3D space. The information originates and end in the digital world.
	Transportation	self-driving cars, drones, autonomous vehicles of any kind, eVTOLs, and also space exploration	Transportation moves people or cargo in the real world.

	Artificial Intelligence (AI)		Artificial intelligence automates the processing of information in the digital world.
	Robots		Robots similarly automate things but they have a “body” and operate in the real world.
	Computing	quantum computing, brain-computer interfaces (BCI)	Computing work in the digital world. In an abstract sense it's the equivalent of energy and power source in the real, physical world.
	Crypto-currency	anything related like blockchain, decentralized finance (DeFi), decentralized autonomous organizations (DAO), non-fungible tokens (NFT)	Crypto-currency, blockchain and distributed ledger technologies work primarily in the digital world. Of course, digital money and DAO impacts the real world too.
	Clean energy	solar power, wind power, geothermal power, tidal or wave energy or anything else that is truly renewable also includes battery technology	Clean energy is the power in the real world.

Solutions can use multiple emerging technologies. The hexagonal cards can be placed next to each other with similar color sides, showing the transitions from the real world to the digital and back.

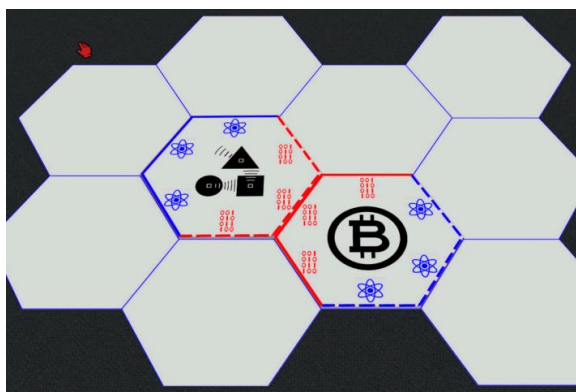


Figure 2 - An "oracle" sends data from the outside world to a blockchain

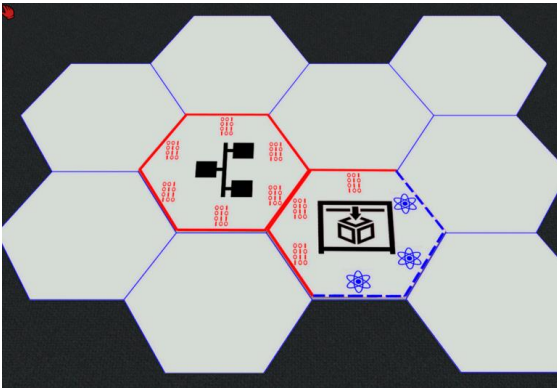


Figure 3 - Distributing design over the Internet and then manufacturing it using 3D Printing

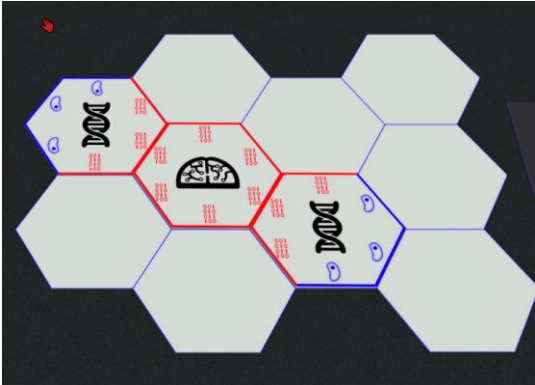


Figure 4 - GMO using AI

Game Play – step-by-step

Choose who will be the dealer, the “master of ceremony”.

Select the order of players, who will go first, who goes second etc.

Setup at the start of the Game

1. The dealer deals everyone five hexagonal emTech cards.

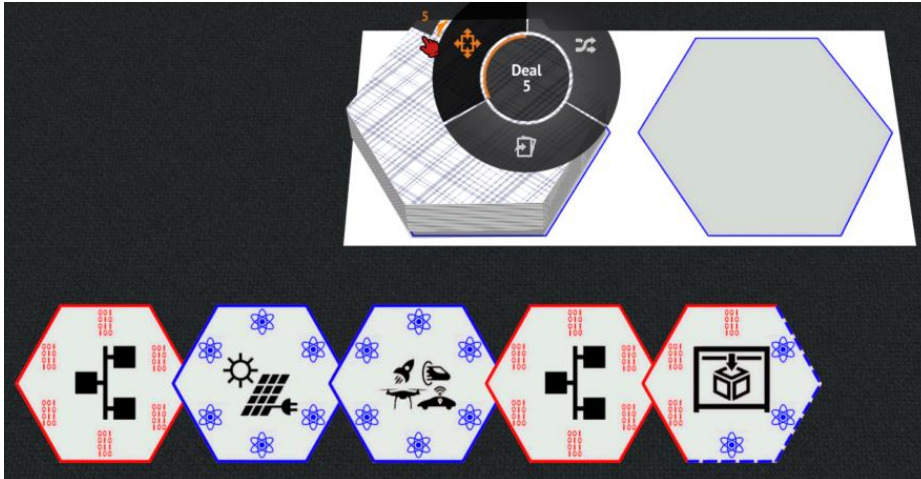


Figure 5 – Five emTech Cards in the Hand

2. The dealer takes as many problem cards as there are players and places them face up onto the pathway boards.

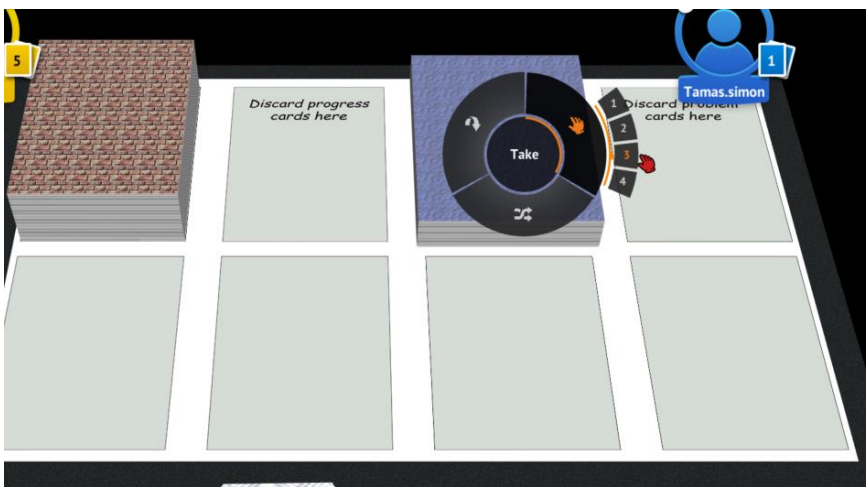


Figure 6 – Take Problem Cards...

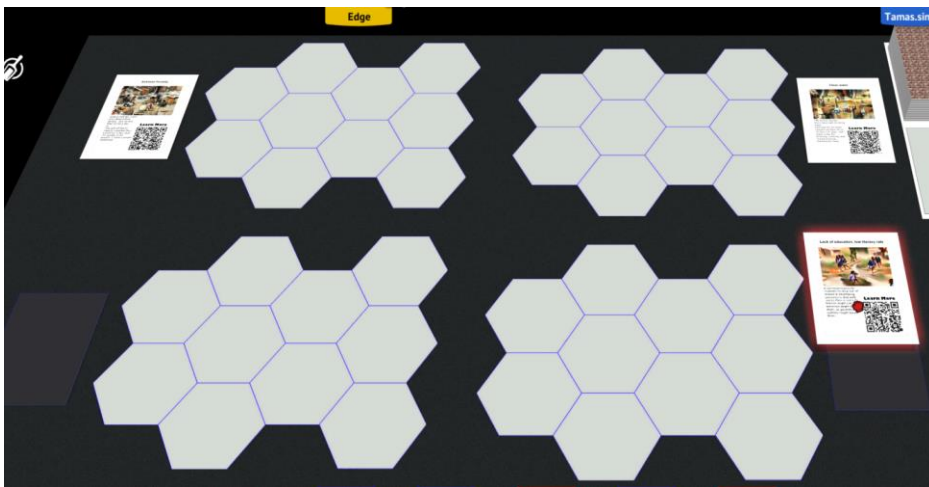


Figure 7 - ...and place them next to the Pathway boards (with the hexa tiles)

Take and place as many Progress cards face up as there are players.

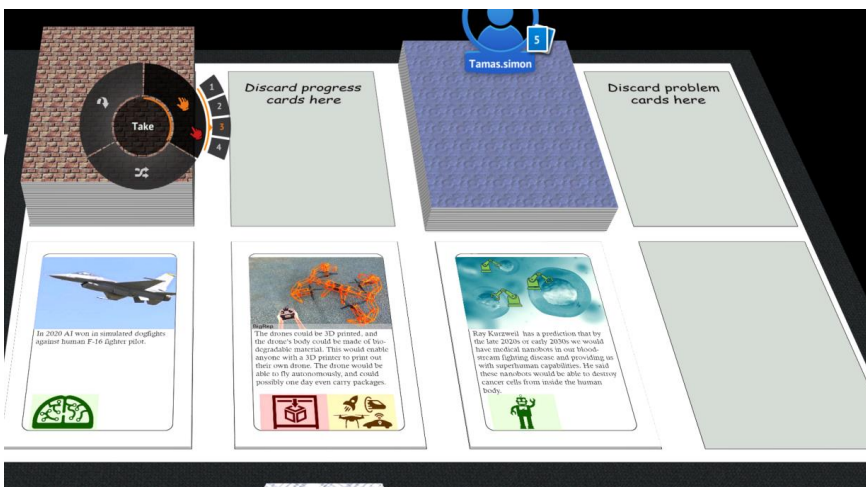


Figure 8 - Progress cards placed face up

Now you are ready to start the first round...

A round

Let the players each take their turn and at the end the dealer will evaluate the impact and hand out any rewards.

A player's turn

Make sure there are 5 cards in the hand

Draw new cards if necessary.

Contribute an idea that helps

The core part of the game is for players to come up with ideas for the application of emTechs to solve grand, global problems. Most solutions will require a combination of multiple emTechs. Here players can work together and build on each-other's ideas.

Players are encouraged to

- Use the face-up progress cards (+1 I\$)
- Use as multiple emTechs
- Build on each other's ideas
- help each other

Voting

Players present their pathways then vote on each others' ideas. It's a form of I-Cut-You-Choose.

Players have one vote each and they **cannot vote on their own idea**.

- An idea with the most votes wins and gets "Philanthropist support". This player wins an extra 5 I\$.
- If there is a draw, then there is no winner.
- Ideas with no votes at all are deemed unreasonable, not feasible and are discarded.

3 Players Voting Results Scenarios

2-1-0

- the one with 2 votes is the winner and gets Philanthropist support
- the one with 1 vote is a viable idea
- the one with no votes is deemed unreasonable, not feasible

1-1-1

- all ideas are viable
- there is no winner

4 Players Voting Results Scenarios

3-1-0-0

- the idea with 3 votes is the winner and gets Philanthropist support
- the idea with 1 vote is a viable idea
- the other ideas with no votes are deemed unreasonable, not feasible

2-2-0-0

- both ideas with 2 votes are viable but neither is winner
- the other two ideas with no votes are deemed unreasonable, not feasible

2-1-1-0

- the idea with 2 votes is the winner and gets Philanthropist support
- the other two ideas with 1 vote each are viable
- the last idea with no votes is deemed unreasonable, not feasible

1-1-1-1

- all ideas are viable
- there is no winner

You can use the little green cubes to cast a vote.

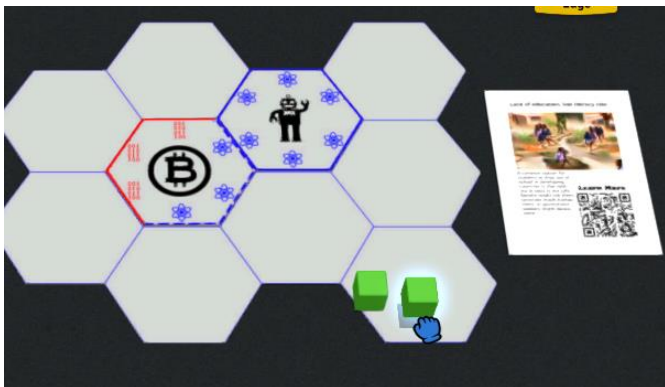


Figure 9 - voting by Placing Green Cubes

End of the round - Evaluating impact

When all players have finished their turns (and before the next round) the dealer or “master of ceremony” – shall evaluate the impact.

Players get rewarded for the positive impact of their accepted ideas. They receive 1 I\$ for each emerging technology hex card used.

Players pay 1I\$ penalty for every problem that is not solved.

Take the problem cards off the table and replace them with new ones drawn.

There is a catch, however. **Existential threats** will not just go away. They remain on the table. These problems can get so bad that if there is no solution found in the subsequent turn either then it's *game over* for everyone!

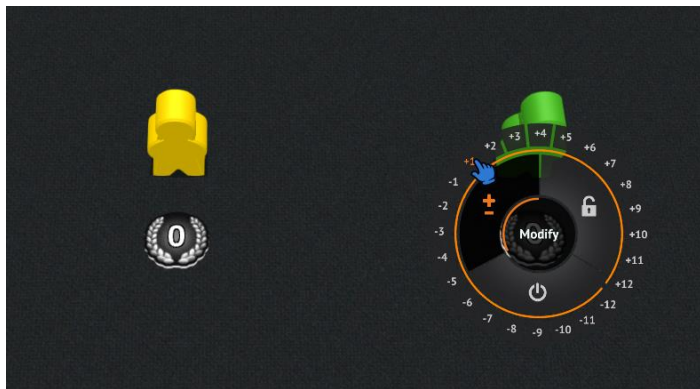


Figure 10 - Use the counters to keep track of I\$s

Summary of rewards:

Description	I\$ reward
Hexagonal emTech card successfully used in an accepted idea	+1 I\$
The emerging Technology on a successfully used card matches one of the face-up Progress cards	+1 I\$
Player won the voting	+ 5 I\$
For each problem left without solution	-1 I\$

Existential Threats – “game over”

The existential threats in the game are

- Rouge superintelligence
- Ecological crises
- Asteroid hitting Earth
- Nuclear weapons in the wrong hands or in an accident
- Climate Crises
- Chemical and Biological Weapons in the wrong hands or in an accident



Cleanup

Clear the table, place any hexagonal emTech cards at the bottom of the deck.

Discard any Progress card that was used and replace them with newly drawn cards placed face up again.

End of Game

The game ends after the end of a round – play as many rounds as you wish.

If players managed to deal with any existential threats, then the player who has the most Impact Dollars at the end wins.

